Surname	Centre Number	Candidate Number
First name(s)		2

GCE A LEVEL

1300U40-1

S25-1300U40-1

THURSDAY, 12 JUNE 2025 – AFTERNOON

MATHEMATICS – A2 unit 4
APPLIED MATHEMATICS B

1 hour 45 minutes

For Examiner's use only		
Question	**Maximum Mark**	**Mark Awarded**
1	**4**	
2	**12**	
3	**8**	
4	**16**	
5	**7**	
6	**9**	
7	**8**	
8	**7**	
9	**9**	
Total	**80**	

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ADDITIONAL MATERIALS

In addition to this examination paper, you will need:

- a Formula Booklet;
- a calculator;
- statistical tables (RND/WJEC Publications).

INSTRUCTIONS TO CANDIDATES

Use black ink or black ball-point pen. Do not use gel pen or correction fluid.

You may use a pencil for graphs and diagrams only.

Write your name, centre number and candidate number in the spaces at the top of this page.

Answer **all** questions.

Take g as $9{\cdot}8\,\mathrm{ms}^{-2}$.

Write your answers in the spaces provided in this booklet. If you run out of space, use the additional page(s) at the back of the booklet, taking care to number the question(s) correctly.

INFORMATION FOR CANDIDATES

The maximum mark for this paper is 80.

The number of marks is given in brackets at the end of each question or part-question.

Sufficient working must be shown to demonstrate the **mathematical** method employed.

Answers without working may not gain full credit.

Unless the degree of accuracy is stated in the question, answers should be rounded appropriately.

You are reminded of the necessity for good English and orderly presentation in your answers.

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Reminder: Sufficient working must be shown to demonstrate the **mathematical** method employed.

Section A: Statistics

1. A continuous random variable X is uniformly distributed over the interval *[c,d]*. The mean and variance of X are 7.5 and 3 respectively. Find the values of *c* and *d*. [4]

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2. Selina is sitting two mutiple-choice tests.

(a) For each question in the first test, she classifies herself as 'very confident', 'confident' or 'not confident' with probabilities 0.2 , 0.5 and 0.3 respectively.

If she is 'very confident', the probability of getting a correct answer is 0.9.

If she is 'confident', the probability of getting a correct answer is 0.7.

If she is 'not confident', the probability of getting a correct answer is 0.25.

i) Calculate the probability that she will get the correct answer to a randomly selected question. [3]

ii) Given that Selina gets a correct answer to randomly selected question, calculate the probability that she was 'not confident'. [2]

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(b) The second test has 10 questions, each with 4 choices from which to select the answer. The probability that Selina knows the answer to a question is p, independently of any other question. If she knows the answer, she will select it. Otherwise, she will select one of the four choices at random, with equal probability for each.

i) Calculate the smallest value of p that will give a probability of at least 0.7 of Selina getting the correct answer to a randomly selected question. [4]

ii) At least 8 out of the 10 questions must be answered correctly to pass the test. Assuming that the probability of Selina getting the correct answer to a randomly selected question is 0.7, calculate the probability that she will pass the test. [3]

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3. A city councilor is investigating the decline in the number of bus drivers in the city. During her research, she finds an article calming that delivery companies are actively recrulting bus drivers to work for them.

She collects data on the number of bus drivers and the number of delivery drivers from 8 randomly selected cities of a similar size to her city. She calculates the correlation coefficient in order to carry out a hypothesis test

(a) Explain why a one-tailed test could be appropriate. [1]

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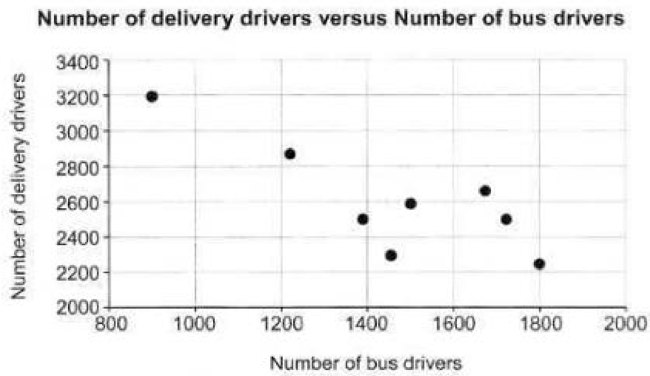

The product moment correlation coefficient for the data is --0.8206.

(b) Carry out a one-tailed significance test at the 5% significance level. [5]

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(b) (i) Suggest a reason why the city councilor might conclude that there is a casual link between the number of delivery drivers and the number of bus drivers.

(b) (ii) What can the city councilor conclude about the recruitment of bus drivers by delivery companies over time?

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4. Collectible football cards come in packs. The masses, in grams of the packs are normally distributed with the mean 28.3 and standard deviation 0.3.

(a) Calculate the probability that a selected pack will have a mass less than 28g. [2]

(b) Packs have a mass more than 28.8g are classed as 'heavy'. Andrei buys a randomly selected 'heavy' pack of cards.

Calculate the probability that this pack has a mass more than 29.1g. [6]

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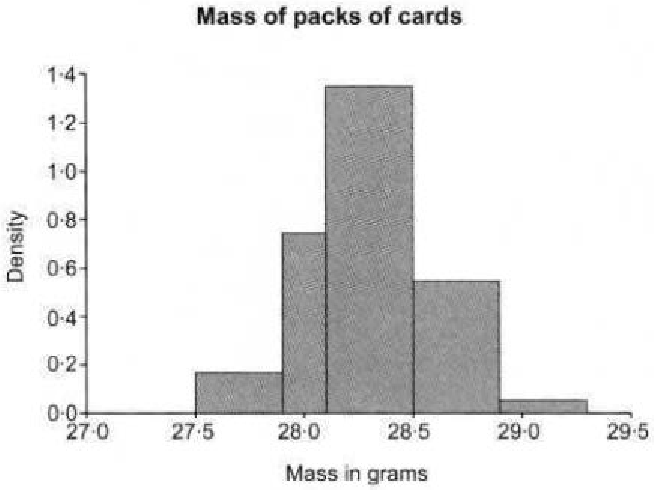
(c) Graham has a large number of packs of cards and wants to sell some to Andrei. Graham claims that the packs that he has to sell are, on average, heavier than 28.3g.

Graham draws a histogram of the masses of his packs of cards, shown below.



Andrei randomly selects 10 packs from Graham to test Graham's claim. He finds that the total masses is 285g.

(i) What feature of the histogram supports conducting a significance test on the sample mean using Andrei's data? [1]

(ii) Given that the population standard deviation of the masses of Graham's pack is 0.32g, test Graham's claim at the 3% level of significance. [7]

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Section B: Differential Equations and Mechanics

5. The diagram below shows a set of traffic lights suspended in equilibrium by means of two cables *AC* and *BC*, inclined at angles of 60° and 45° to the vertical respectively. The set of traffic lights has a mass of 25kg and may be modeled as a single particle attached at C.

(a) Show that the tension *AC* is approximately 179N, and calculate the tension in BC. [6]

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(b) State a modeling assumption that you have made in your conclusion. [1]

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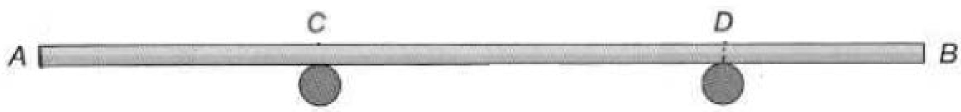
6. The diagram below shows a unifrom rod *AB*, of lengh 3.8m and weight 180N. The rod is resting horizontal in equilibrium on two smooth supports at the points C and D, where DB = 0.8m and CD = 1.8m



An object of weight *W* N is placed at *A*.

(a) Given that W=69, find the magnitudes of the reaction at C and the reaction at D [6]

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(b) Determine the greatest value of W for which the rod remains in equilibrium. [3]

7. A particle of mass 1.2kg is moving in a straight line under the action of a variable force FN,

$$F = \frac{30}{(t+2)^2}$$

The speed of the particle at time t seconds is v ms^{-1},
When $t = 0.5$, the speed of the particle is 20 ms^{-1}.

(a) Find an expression for v in terms of t and hence write down the limiting value of v as t increases. [6]

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(a) Calculate the time taken for particle to reach a speed of 29 ms^{-1}. [2]

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8. An object, of mass 5kg, is projected up the line of greatest slope of a rough plane inclined at an angle of 30^{o} to the horizontal.

(a) Given that the deceleration of the object up the plane is 6.3 ms^{-2}, show that the coefficient of friction between the plane and the object is $\dfrac{2\sqrt{3}}{21}$ [5]

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(b) Determine whether the object will remain stationary on the plane immediately after after it comes to rest. [2]

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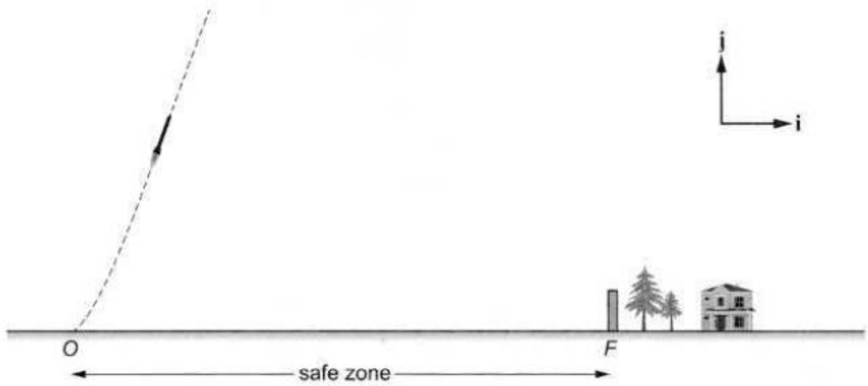
9. A small rocket R is launched from a point O on horizontal ground so that it is moving in a vertical plane, where the unit vectors **i** and **j** are horizontal and vertical respectively.

The diagram below shows the path of R for the first 4 seconds of its motion.

The rocket R is required to land in a 'safe zone' which consist of the horizontal line from O to F.



At time t seconds, the position vector **r** m of R, relative to O, is modeled by

$$\mathbf{r} = 4t^2\mathbf{i} + \left(\frac{1}{4}t^4 + 6t^{\frac{3}{2}}\right)\mathbf{j} \qquad \text{for} \qquad 0 \leqslant t \leqslant 4.$$

(a) Find an expression for the velocity vector of R at time ts, valid for $0 \leq t \leq 4$. Hence find the velocity vector of R when $t = 4$.

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(b) At $t = 4$, R runs out of fuel. For $t > 4$, the motion of R can be modeled as a projectile.

(i) Show that R takes approximately 18 seconds to reach the ground after the fuel runs out, [4]

(ii) Hence, determine whether R lands within the 'safe zone', given that $OF = 600$m. [2]

END OF PAPER

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